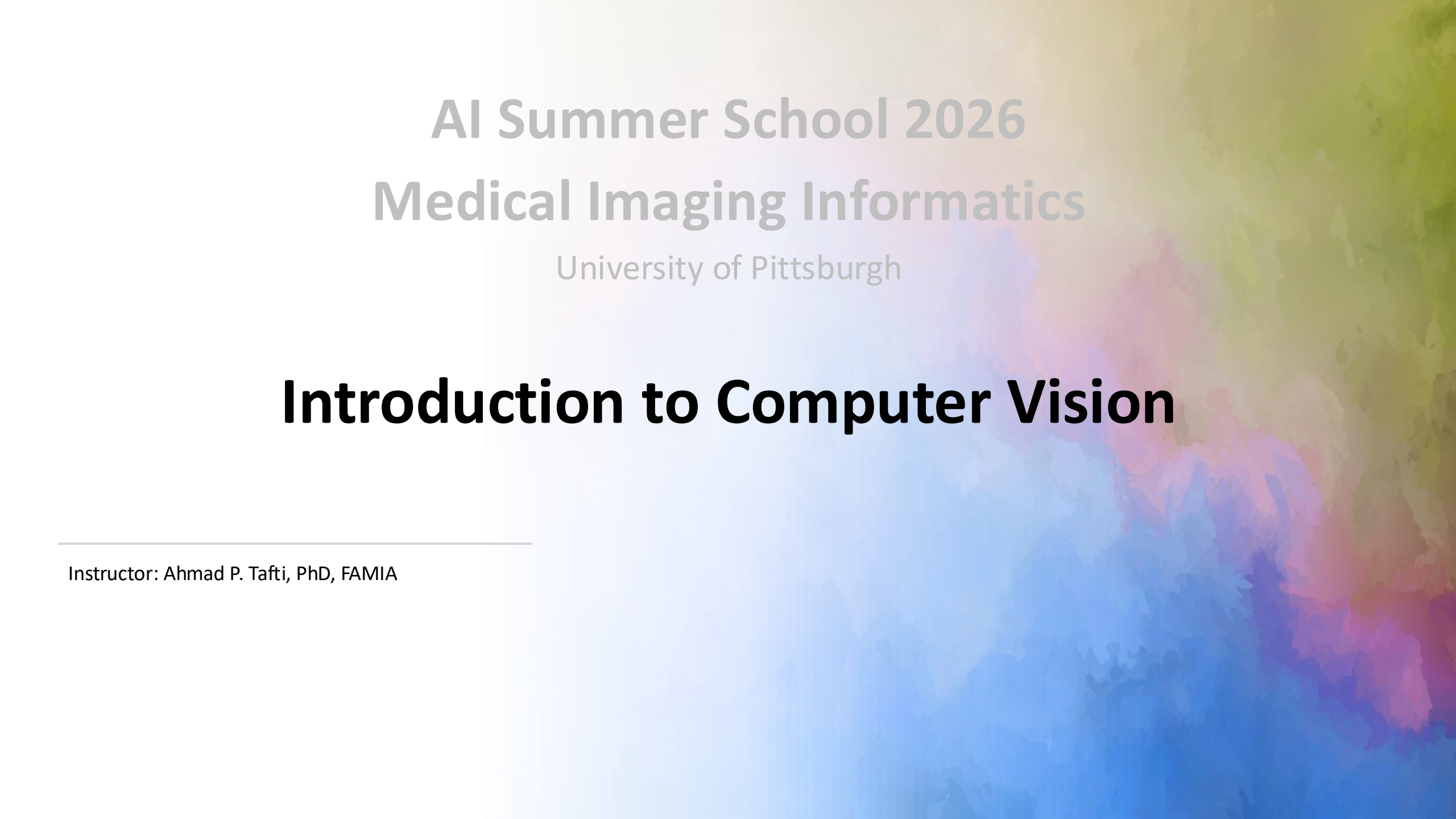




AI Summer School 2026

Medical Imaging Informatics

University of Pittsburgh

Introduction to Computer Vision

Instructor: Ahmad P. Tafti, PhD, FAMIA

Learning Objectives

After completing this lecture, you should be able to:

- Understand and explain computer vision briefly
- Explain the motivation of computer vision
- Explain computer vision/imaging informatics application in real-world scenarios
- Discuss computer vision algorithms

Outline

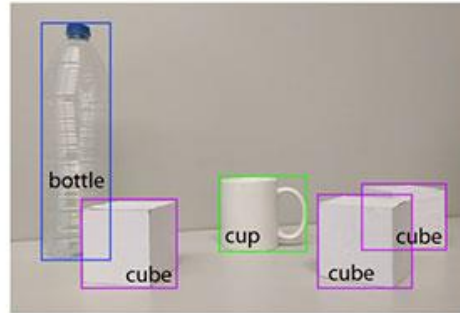
- Computer Vision/Imaging Informatics; What and Why?
- Computer Vision Applications
- Computer Vision Algorithms; Big Picture

Computer Vision | Imaging Informatics

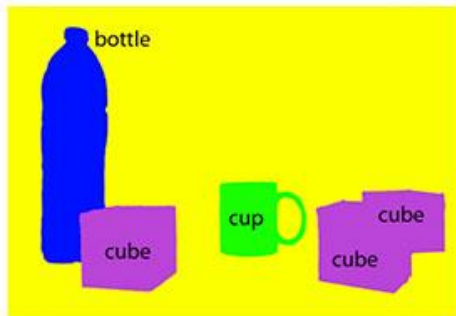
- Computer vision deals with how computers can understand from digital images and video sequences.



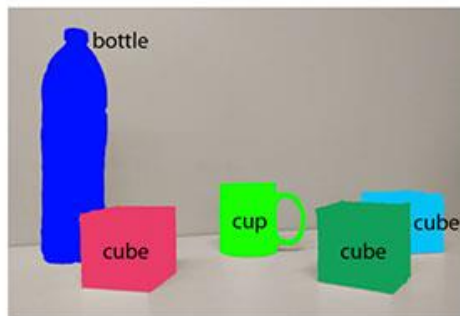
(a) Image classification



(b) Object localization



(c) Semantic segmentation



(d) Instance segmentation

Object Localization and Semantic Segmentation

<https://www.pyimagesearch.com/>



3D Surface Reconstruction

Motivation: Sight (Vision), Sound (Hearing), Smell, Taste, and Touch

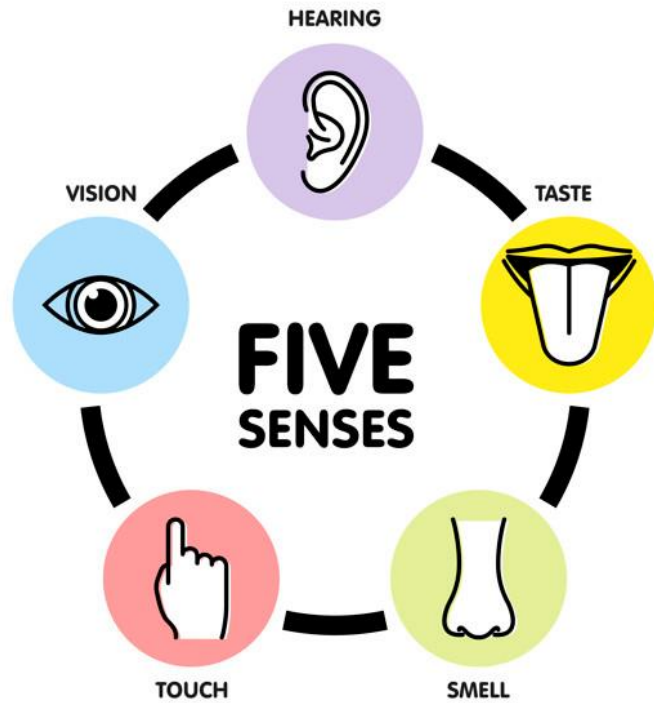
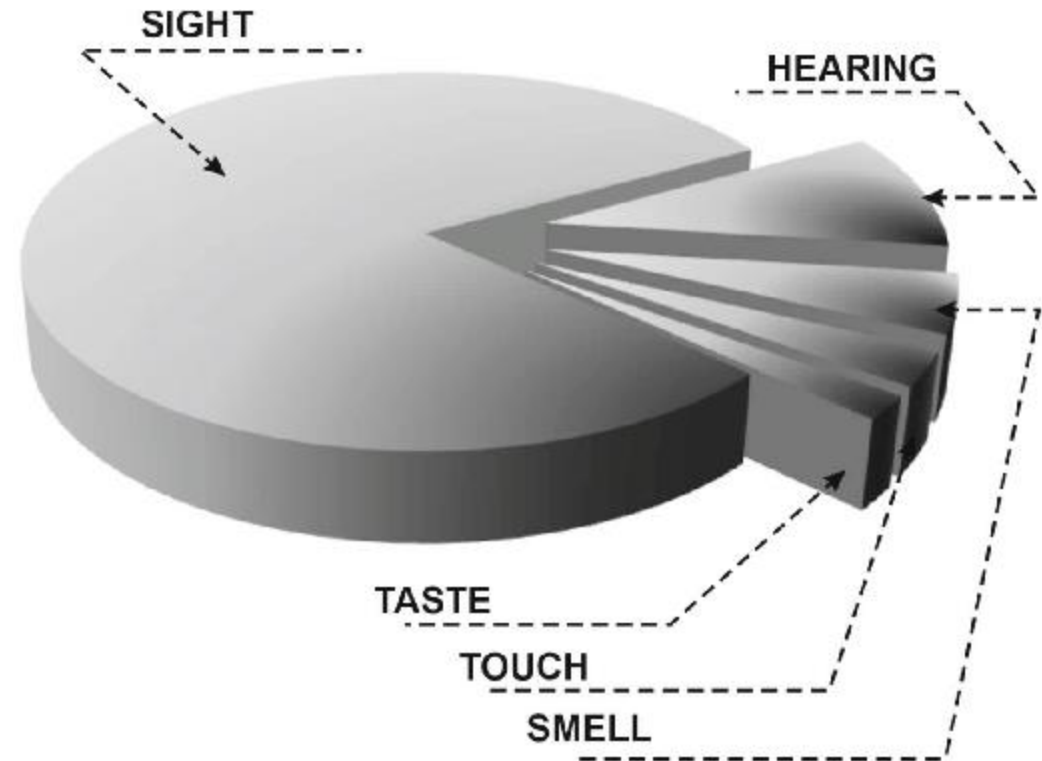
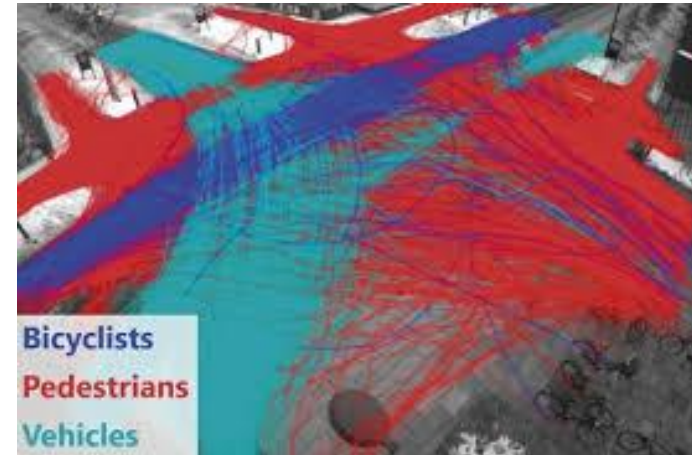


Image from: <https://www.freepik.com>

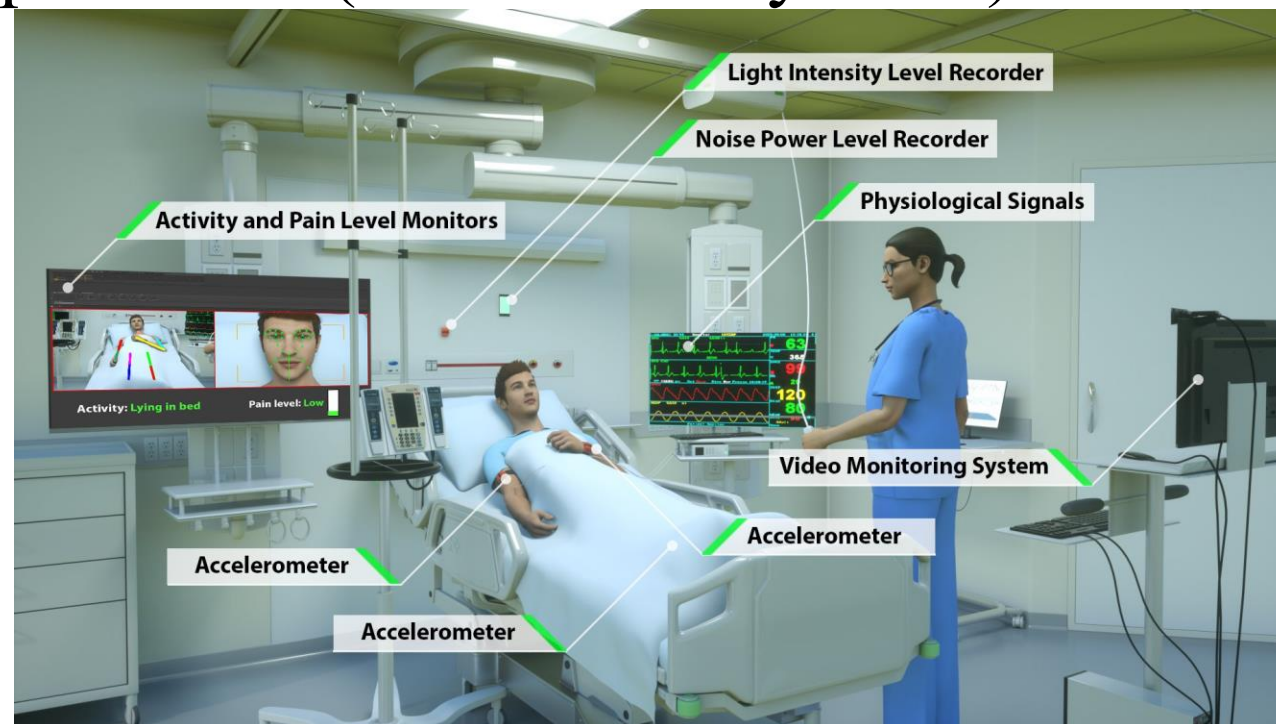
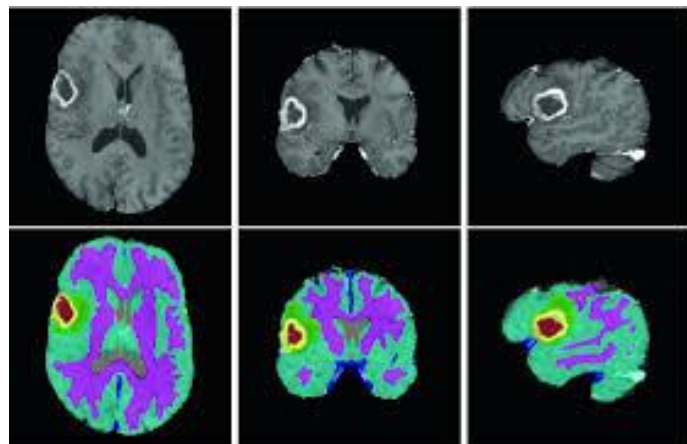


Citation: Application of sEMG and Posturography as Tools in the Analysis of Biosignals of Aging Process of Subjects in the Post-production Age

Motivation: Computer Vision Applications (Surveillance Systems)



Motivation: Computer Vision Applications (Healthcare Systems)



Motivation: Computer Vision Applications (QA/QC Systems)

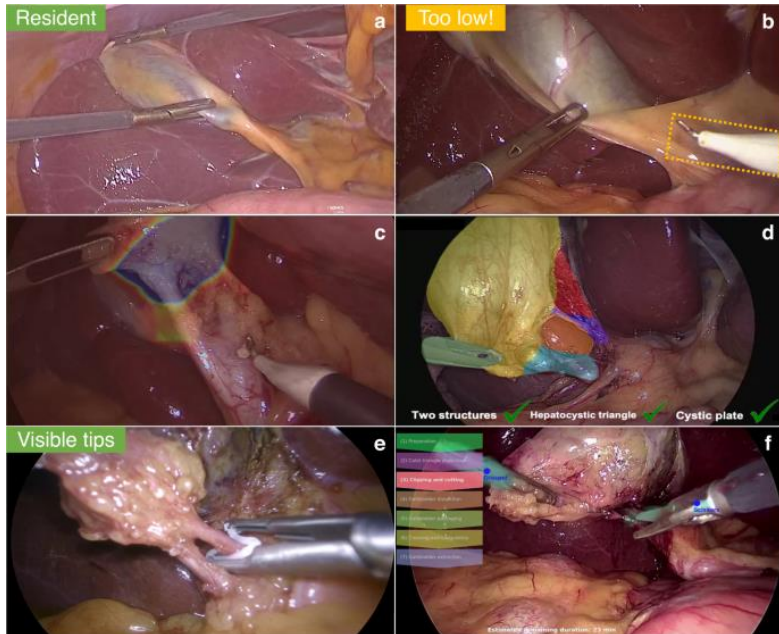


Motivation: Computer Vision Applications (Robotics)

Motivation: Computer Vision Applications (Self Driving Cars)



Motivation: Computer Vision Applications (Robot Assisted Surgery)



Successes with Computer Vision

- Optical character recognition (OCR)
- Retail (e.g., automated checkouts)
- 3D model building (and 3D printing)
- Medical imaging
- Automotive safety
- Surveillance
- Fingerprint recognition and biometrics

“Computer Vision”

- Machine Vision
- Robot Vision
- Image Analysis
- Image Processing
- Digital Image Processing
- Imaging Informatics

Video Analysis

How Computer Vision Works?

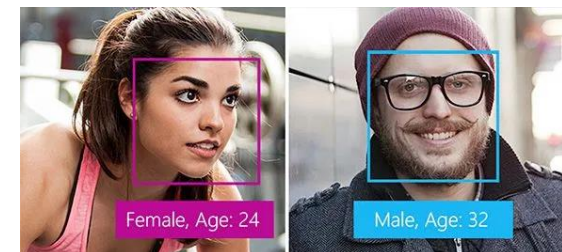
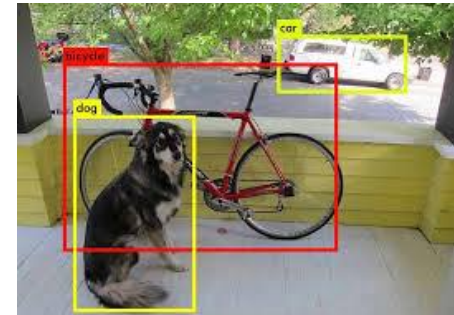
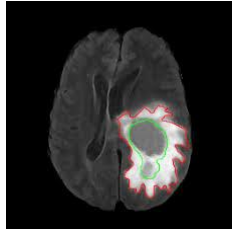
- 1) Acquiring an image
- 2) Processing Image
- 3) Understanding Image



Computer Vision Mechanisms/Algorithms

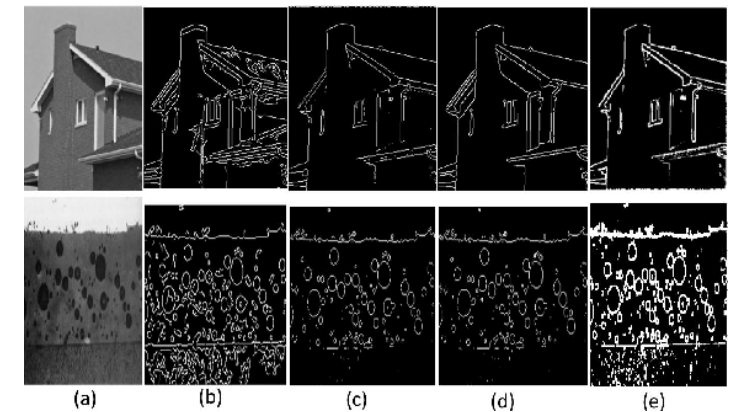
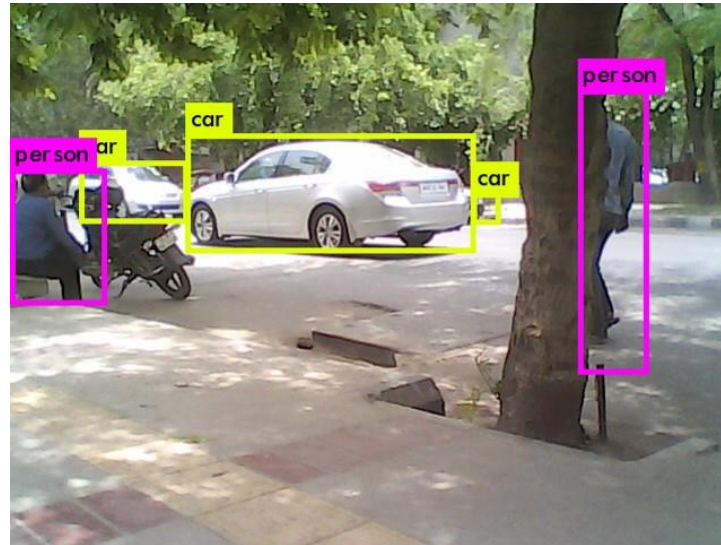
There are **many types of computer vision algorithms** that are used in different ways:

- **Image segmentation:** partitions an image into multiple regions to be examined separately.
- **Object detection:** identifies a specific object in an image. Advanced object detection recognizes many objects in a single image
- **Facial recognition:** it is an advanced type of object detection that not only recognizes a human face in an image but can also identify a specific individual.



Computer Vision Mechanisms/Algorithms

- **Edge detection:** is a technique used to identify the outside edge of an object or landscape to better identify what is in the image.
- **Pattern detection:** is a process of recognizing repeated shapes, colors and other visual indicators in images.
- **Image classification:** groups images into different categories.
- **Feature matching:** is a type of pattern detection that matches similarities in images to help classify them.



Low-Level, Mid-Level, and High-Level Image Processing

Type	Input	Output	Examples
Low Level Process	Image	Image	Noise removal, image sharpening
Mid-Level Process	Image	Attributes	Object recognition, Segmentation
High Level Process	Attributes	Understanding	Scene understanding, autonomous navigation

Thank you!

Questions!



GitHub



Pitt AI Summer School